

Ashok Reddy Bhimavarapu

Data Analytics, Data Engineering and Machine Learning | SQL, Python, dbt, Snowflake, Power BI, LLMs

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PROFESSIONAL SUMMARY

Data professional with 4+ years of experience across analytics, data engineering and machine learning in financial services, healthcare and technology, and an MS in Information Science (AI and Machine Learning). Builds the full path from raw data to decision: pipelines and tested data models in dbt and Snowflake, dashboards and KPI frameworks in Power BI and Tableau, and forecasting and classification models in Python. Known for making numbers trustworthy, with validation, reconciliation and one definition per metric, and for explaining results clearly to non-technical leaders.

TECHNICAL SKILLS

- **Languages:** SQL (advanced), Python (pandas, NumPy, scikit-learn, statsmodels, pytest), R, JavaScript (React)
- **Data Engineering:** dbt (incremental models, macros, tests, docs), ETL/ELT, dimensional modeling, Snowflake, BigQuery, Redshift, PostgreSQL, SQL Server, Oracle, AWS (S3, Glue, SageMaker), GCP, Azure, Git, CI/CD, FastAPI, REST APIs
- **Analytics and BI:** Power BI (DAX, Power Query), Tableau, Excel (pivot tables, lookups, VBA), KPI frameworks, A/B testing, forecasting, cohort and funnel analysis, root-cause analysis, dashboards, data storytelling
- **Machine Learning and AI:** Gradient boosting (XGBoost), regression, classification, clustering, time-series forecasting (ARIMA, LSTM), anomaly detection, NLP, Keras, LLM evaluation, prompt design, data preparation for RAG
- **Data Quality and Governance:** Validation frameworks, reconciliation, data quality monitoring and alerts, documentation, lineage, access controls
- **Ways of Working:** Requirements gathering, stakeholder communication, Agile/Scrum, Jira

PROFESSIONAL EXPERIENCE

Data Analyst | Broadridge Financial Solutions, Inc. | New York, NY | Feb 2025 – Present

- Own dbt models in Snowflake over transaction-processing data at billions-of-rows scale, covering design, transformation logic, testing and documentation, with staging, intermediate and mart layers, incremental models, Jinja macros and custom tests.
- Designed and built from scratch an automated SQL validation framework that reconciles three source systems before every delivery, cutting recurring manual review effort by 25–30%.
- Built and validated 10–15 SQL and dbt models that supply an internal RAG assistant with transaction, client, billing and reference data: standardised definitions, added column descriptions, flattened joins and restricted exposed fields. Reconciliation checks caught duplicate records inflating a reported metric before release.
- Build Power BI and Tableau dashboards and a KPI framework and scorecards for directors, with one definition and one source per metric, and manage the team's BI environment, including workspaces, access and refreshes.
- Automate reporting and validation workflows with Python, SQL and spreadsheet macros, saving the team 5–8 hours each week, and set up alerts for SLA breaches and data quality failures.
- Build machine learning models in Python for forecasting, risk and anomaly detection, each measured against a test set before release.
- Tune Snowflake performance with clustering keys, warehouse sizing and query optimization, parse semi-structured JSON, and manage changes through Git pull requests, code review and CI/CD.
- Support pricing, cost and margin analysis for client programs, monthly forecasting, annual planning and executive reporting packages.
- Trace data issues to their root cause with upstream engineering teams, and present findings and recommendations to non-technical leaders.

Data Analyst | Optum Global Solutions | Gurugram, India | Nov 2022 – Dec 2023

- Built and loaded claims, member, provider and clinical tables in Amazon Redshift with scheduled SQL-based ETL from AWS, working with ICD-10, CPT/HCPCS and NDC coded data.
- Rebuilt manually assembled reporting as SQL-driven dashboards, bringing turnaround from 1–2 days to same-day.
- Built what-if forecasting models in Python and Excel that linked claims volume to staffing and were used to plan capacity.
- Developed NLP and machine learning models in Python on free-text clinical notes and claims data.
- Checked data for missing values, duplicates and claim discrepancies, and ran cost and pricing analysis on claims and provider data for business and clinical operations stakeholders.

Data Analyst | Dell Technologies | Hyderabad, India | Apr 2021 – Oct 2022

- Built and scheduled SQL transformations in Google BigQuery into daily-refreshed, partitioned and clustered tables from sales, order and CRM data.
- Built Power BI dashboards for customer-facing regional sales teams, and reconciled CRM, sales and operational data across AWS and GCP sources.
- Produced bill of materials cost rollups and bi-weekly product cost reports from SAP, working with Sourcing and PLM teams to verify changes.
- Ran pricing and margin analysis and monthly demand forecasting for sales teams.
- Automated weekly reports with Python, SQL and spreadsheet macros.

PROJECTS

Fund Proxy Voting Analysis | Python, DuckDB, dbt, SQL, LLM evaluation, GitHub Actions

- Built a pipeline over 73.8 million fund votes from 33,929 SEC filings (12,416 funds), with dbt models that standardise 40+ spellings of vote values and 12 date formats, covered by 21 data tests and CI.
- Matched 8,274 proposal wordings into 3,850 proposals and labelled their direction using two independent signals; found ESG-named funds backed 33% of specialist-supported proposals against 14% for other funds.
- Evaluated an LLM (gpt-oss-120b via the Groq API) for labelling proposal direction against an 815-item answer key; prompt design raised held-out accuracy from 69% to 85%, and a rules-plus-LLM hybrid reached 91% on 578 held-out proposals.

Olist Analytics Engineering | dbt, DuckDB, SQL, Python, Power BI, GitHub Actions

- Modelled 10 raw e-commerce and marketing tables into staging, intermediate and mart layers with dbt (24 models, including an incremental orders table), covered by 80 tests that run in GitHub Actions on every push.
- Built a Power BI dashboard and a lead-scoring model on the marts; on the most recent 2,000 leads, the top 10% of scores closed at 23% against a 10.5% average.

Stock Market Analysis and Prediction Web App | Python, FastAPI, React, SQLite

- Built a full-stack web app that shows 7-day price forecasts alongside sentiment analysis of financial news for NASDAQ and NSE stocks, with a FastAPI back end, a React front end and a SQLite database.
- Trained and compared three forecasting approaches (ARIMA, LSTM and linear regression) on five years of daily price data, and combined the forecast trend with news sentiment to produce a buy, sell or hold signal.

EDUCATION

Master of Science, Information Science (Artificial Intelligence and Machine Learning) | University at Albany, SUNY | Albany, NY | 2024 – 2025 | GPA 3.71/4.00

Bachelor of Technology, Computer Science and Engineering | Sathyabama University | Chennai, India | 2018 – 2022 | GPA 7.91/10.00

PUBLICATIONS

- Employee Performance Estimation Using Machine Learning, IJSDR (IJSDR2304246), April 2023. Random Forest classification model on 5,500 HR records and 31 features.